**DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS**

# Mathematics II Advanced Calculus (2AC)

**Exercise Sheet 3 (for your exam practice):** **June 2023**

**Solutions:**

1. Let $\mathcal{C}$ be the curve described by the vector function $\vec{r}: \big[-\pi\,,\ \frac{5\pi}{8}\big] \to \mathbb{R}^3$ defined by

$$\vec{r}(t) := \Big(2\sin(t-\pi)+\frac{\pi}{2},\ \cos(\pi+t),\ -\sqrt{3}\sin\big(\frac{\pi}{2}-t\big)\Big), \quad -\pi \le t \le \frac{5\pi}{8}\,.$$

(a) Show that $\mathcal{C}$ is a regular curve.

We find that

$$\vec{r}\,'(t) = \Big(2\cos(t-\pi),\ -\sin(\pi+t),\ \sqrt{3}\cos\big(\frac{\pi}{2}-t\big)\Big) \quad \text{continuous}\,.$$

Now, if there is $t \in \big(-\pi\,,\ \frac{5\pi}{8}\big)$ such that $\vec{r}\,'(t) = (0,0,0)$, then we get

$$\cos(t-\pi) = 0, \qquad \sin(\pi+t) = 0, \qquad \cos\big(\frac{\pi}{2}-t\big) = 0\,.$$

Noticing that $\cos(t-\pi) = -\cos t$ and $\sin(\pi+t) = -\sin t$ we then that $\cos t = \sin t = 0$ which is impossible. Thus, $\vec{r}\,'(t) \neq (0,0,0)$, for every $t \in \big(-\pi\,,\ \frac{5\pi}{8}\big)$ and therefore, the curve $\mathcal{C}$ is regular.

(b) Show that $\vec{r}\,'(t) = \big(-2\cos t, \sin t,\ \sqrt{3}\sin t)\big)$.

We just notice from above that

$$\cos(t-\pi) = -\cos t, \quad \sin(\pi+t) = -\sin t \quad \text{and} \quad \cos\big(\frac{\pi}{2}-t\big) = \sin t$$

$$\Rightarrow \quad \vec{r}\,'(t) = \Big(2\cos(t-\pi),\ -\sin(\pi+t),\ \sqrt{3}\cos\big(\frac{\pi}{2}-t\big)\Big) = \big(-2\cos t, \sin t,\ \sqrt{3}\sin t\big)\,.$$

(c) Find the curvature $\kappa$ of the curve $\mathcal{C}$ at the point $\big(\frac{\pi}{2}, -1, -\sqrt{3}\big)$.

We first seek $t \in \big[-\pi\,,\ \frac{5\pi}{8}\big]$ such that

$$\vec{r}(t) = \big(\frac{\pi}{2}, -1, -\sqrt{3}\big) \ \Leftrightarrow \ \begin{cases} 2\sin(t-\pi)+\frac{\pi}{2} = \frac{\pi}{2}\,, \\ \cos(\pi+t) = -1, \\ -\sqrt{3}\sin\big(\frac{\pi}{2}-t\big) = -\sqrt{3}, \\ -\pi \le t \le 5\pi/8 \end{cases} \Leftrightarrow \ \begin{cases} \sin(t-\pi) = 0\,, \\ \cos(\pi+t) = -1, \\ \sin\big(\frac{\pi}{2}-t\big) = 1, \\ -\pi \le t \le 5\pi/8 \end{cases} \Leftrightarrow \ t = 0\,.$$

So, we want to find $\kappa(0) = \dfrac{\|\vec{r}\,'(0) \times \vec{r}\,''(0)\|}{\|\vec{r}\,'(0)\|^3}$.

We find that

$$\begin{cases} \vec{r}\,'(0) = (-2,0,0), \\ \vec{r}\,''(t) = \Big(-2\sin(t-\pi),\ -\cos(\pi+t),\ \sqrt{3}\sin\big(\frac{\pi}{2}-t\big)\Big) \Rightarrow \quad \vec{r}\,''(0) = (0,1,\sqrt{3})\,. \end{cases}$$

Hence,

$$\vec{r}\,'(0)\times\vec{r}\,''(0) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -2 & 0 & 0 \\ 0 & 1 & \sqrt{3} \end{vmatrix} = (0, 2\sqrt{3}, -2) \quad \Rightarrow \quad \kappa(0) = \frac{\|\vec{r}\,'(0)\times\vec{r}\,''(0)\|}{\|\vec{r}\,'(0)\|^3} = \frac{4}{8} = \frac{1}{2}.$$

(d) Write down the cartesian equation of the normal plane to the curve $\mathcal{C}$ at the point in (c).

The normal plane has normal vector the unit tangent vector $\vec{T}(0)$ or $\vec{r}\,'(0)$. Hence, the nromal plane has cartesian equation

$$\Big(x-\frac{\pi}{2},\, y+1,\, z+\sqrt{3}\Big)\cdot(-2,0,0) = 0 \quad \Leftrightarrow \quad x = \frac{\pi}{2}.$$

(e) Find the length of the arc of the curve $\mathcal{C}$ from the point $\big(\frac{\pi}{2}, 1, \sqrt{3}\big)$ to $\big(\frac{\pi-4}{2},\, 0,\, 0\big)$.

We first find the values of the parameter $t$ corresponding to the given points. We find $t=-\pi$ for the first point $\big(\frac{\pi}{2}, 1, \sqrt{3}\big)$ and $t=\frac{\pi}{2}$ for the second point $\big(\frac{\pi-4}{2},\, 0,\, 0\big)$. Thus, the length of the arc of the curve $\mathcal{C}$ from the point $\big(\frac{\pi}{2}, 1, \sqrt{3}\big)$ to $\big(\frac{\pi-4}{2},\, 0,\, 0\big)$ is equal to

$$L = \int_{-\pi}^{\frac{\pi}{2}} \|\vec{r}\,'(t)\|dt = \int_{-\pi}^{\frac{\pi}{2}} \sqrt{(-2\cos t)^2 + (\sin t)^2 + (\sqrt{3}\sin t)^2}\, dt = \int_{-\pi}^{\frac{\pi}{2}} 2\, dt = 3\pi.$$

2. Consider the function $f:\mathbb{R}^2\to\mathbb{R}$ defined by

$$f(x,y) := \begin{cases} \dfrac{\sin^4(x)+\sin^4(y)}{x^2+y^2} & \text{if } (x,y)\neq(0,0), \\[2ex] 0 & \text{if } (x,y)=(0,0), \end{cases} \qquad \forall (x,y)\in\mathbb{R}^2.$$

(a) Study the continuity of $f$ at $(0,0)$. (Hint: $|\sin t|\le |t|$ for every $t\in\mathbb{R}$.)

**First solution:** Along the axes, we find that

$$f(x,0) = \frac{\sin^4(x)}{x^3} = \sin x\Big(\frac{\sin x}{x}\Big)^3 \to 0\times 1 = 0.$$

Similarly, $f(0,y)\to 0$. Hence, the candidate limit is $L=0$.

Using the hint, we get $\sin^4(x)\le x^4$ and $\sin^4(y)\le y^4$. Hence,

$$\begin{aligned} 0\le f(x,y) &= \frac{\sin^4(x)+\sin^4(y)}{x^2+y^2} \le \frac{x^4+y^4}{x^2+y^2} = x^2\times\underbrace{\Big[\frac{x^2}{x^2+y^2}\Big]}_{\le 1} + y^2\times\underbrace{\Big[\frac{y^2}{x^2+y^2}\Big]}_{\le 1} \\ &\le x^2+y^2 \longrightarrow 0 \text{ as } (x,y)\to(0,0). \end{aligned}$$

We could also use $\sin^2(x)\le x^2$ and get

$$\begin{aligned} 0\le f(x,y) &= \sin^2(x)\times\frac{\sin^2(x)}{x^2+y^2} + \sin^2(y)\times\frac{\sin^2(y)}{x^2+y^2} \\ &\le \sin^2(x)\times\Big[\frac{x^2}{x^2+y^2}\Big] + \sin^2(y)\times\Big[\frac{y^2}{x^2+y^2}\Big] \le \sin^2(x)+\sin^2(y). \end{aligned}$$

Hence, $\lim_{(x,y)\to(0,0)} f(x,y) = 0$ by the squeeze theorem and therefore, the function $f$ is continuous at the point $(0,0)$.

**Second solution:** Along the axes we find the candidate limit $L = 0$ as above.

$$\begin{cases} x^2+y^2 \geq x^2 \\ x^2+y^2 \geq y^2 \end{cases} \quad \Rightarrow \quad 0 \leq f(x,y) = \frac{\sin^4(x)}{x^2+y^2} + \frac{\sin^4(y)}{x^2+y^2} \leq \frac{\sin^4(x)}{x^2} + \frac{\sin^4(y)}{y^2}$$

$$= \quad \sin^2(x) \times \Big(\frac{\sin x}{x}\Big)^2 + \sin^2(y)\Big(\frac{\sin y}{y}\Big)^2 \to 0 \times 1 + 0 \times 1 = 0\,.$$

Hence, $\lim_{(x,y)\to(0,0)} f(x,y) = 0$ by the squeeze theorem and therefore, the function $f$ is continuous at the point $(0,0)$.

**Third solution:** Along the axes we find the candidate limit $L = 0$ as above.
Using polar coordinates $r, \theta$ through $x = r\cos\theta$ and $y = r\sin\theta$, and using also the hint, we get

$$\begin{aligned} 0 \leq f(r\cos\theta, r\sin\theta) &= \frac{\sin^4(r\cos\theta) + \sin^4(r\sin\theta)}{r^2} \leq \frac{(r\cos\theta)^4 + (r\sin\theta)^4}{r^2} \\ &= r^2(\cos^4(\theta) + \sin^4(\theta)) \leq 2r^2 \longrightarrow 0 \quad \text{as } r \to 0^+\,. \end{aligned}$$

Hence, $\lim_{r\to 0^+} f(r\cos\theta, r\sin\theta) = 0$ uniformly in $\theta$. Thus, $\lim_{(x,y)\to(0,0)} f(x,y) = 0$ by the squeeze theorem and therefore, the function $f$ is continuous at the point $(0,0)$.

(b) Find, if they exist, $f_x(0,0)$ and $f_y(0,0)$.

We find that

$$\lim_{(x\to 0} \frac{f(x,0)-f(0,0)}{x} = \lim_{x\to 0} \frac{\sin^4(x)}{x^3} = \lim_{x\to 0} (\sin x) \times \Big(\frac{\sin x}{x}\Big)^3 = 0 \times 1 = 0\,.$$

Hence, $f_x(0,0) = 0$. Similarly, by symmetry, also $f_y(0,0) = 0$.

(c) Say whether the function $f$ is differentiable at $(0,0)$.

We want to check whether $\displaystyle\lim_{(x,y)\to(0,0)} \frac{f(x,y)-f(0,0)-f_x(0,0)(x-0)-f_y(0,0)(y-0)}{\sqrt{(x-0)^2+(y-0)^2}} = 0.$

**First solution:** Notice that

$$\frac{f(x,y)-f(0,0)-f_x(0,0)(x-0)-f_y(0,0)(y-0)}{\sqrt{(x-0)^2+(y-0)^2}} = \frac{\sin^4(x)+\sin^4(y)}{(x^2+y^2)^{3/2}}$$

$$\begin{aligned} &\leq \frac{x^4+y^4}{(x^2+y^2)^{3/2}} = \frac{x^4}{(x^2+y^2)^{3/2}} + \frac{y^4}{(x^2+y^2)^{3/2}} \leq \frac{(x^2+y^2)^2}{(x^2+y^2)^{3/2}} + \frac{(x^2+y^2)^2}{(x^2+y^2)^{3/2}} \\ &= 2\sqrt{x^2+y^2} \quad \longrightarrow 0 \quad \text{as } (x,y) \to (0,0)\,. \end{aligned}$$

Thus, the function $f$ is differentiable at the point $(0,0)$.

**Second solution:** One could also write

$$\frac{f(x,y)-f(0,0)-f_x(0,0)(x-0)-f_y(0,0)(y-0)}{\sqrt{(x-0)^2+(y-0)^2}} = \frac{\sin^4(x)+\sin^4(y)}{(x^2+y^2)^{3/2}}$$

$$\begin{aligned} &= \frac{\sin^4(x)}{(x^2+y^2)^{3/2}} + \frac{\sin^4 y}{(x^2+y^2)^{3/2}} \leq \frac{\sin^4 x}{|x|^3} + \frac{\sin^4 y}{|y|^3} \\ &= |\sin x| \times \Big|\frac{\sin x}{x}\Big| + |\sin y| \times \Big|\frac{\sin y}{y}\Big| \to 0 \times 1 + 0 \times 1 = 0\,. \end{aligned}$$

Thus, the function $f$ is differentiable at the point $(0,0)$.

3. Let $S := \{(x,y,z) \in \mathbb{R}^3\colon\ -x^2y + z^2 + y - yz = 1\}$. Find the point(s) $(x,y,z)$ on the surface $S$ at which the tangent plane is parallel to the $xz$-plane..

Let $f(x,y,z) = -x^2y + z^2 + y - yz$. The tangent plane to the surface $\mathcal{S}_1$ at the point $(x,y,z)$ has normal vector $\nabla f(x,y,z)$. Notice that

$$\nabla f(x,y,z) = (-2xy,\ -x^2 + 1 - z,\ 2z - y).$$

The tangent plane to the surface $\mathcal{S}_1$ at a point $(x, yz)$ is parallel to the $xz$-plan if and only if $\nabla f(x,y,z)$ is parallel to the vector $\vec{j} = (0,1,0)$. That is, we want find points $(x,y,z) \in \mathcal{S}_1$ such that

$$\begin{cases} \nabla f(x,y,z) // \vec{j} \\ f(x,y,z) = 1 \end{cases} \Leftrightarrow \begin{cases} f_x(x,y,z) = 0, \\ f_z(x,y,z) = 0, \\ f(x,y,z) = 1. \end{cases} \Leftrightarrow \begin{cases} -2xy = 0, \\ 2z - y = 0, \\ -x^2y + z^2 + y - yz = 1. \end{cases}$$

From the first equation, we get $x = 0$ or $y = 0$.

If $x = 0$ we get $\begin{cases} x = 0, \\ y = 2z, \\ z^2 + y - yz = 1, \end{cases} \Rightarrow \begin{cases} x = 0, \\ y = 2z, \\ z^2 - 2z + 1 = 0, \end{cases} \Rightarrow \boxed{(0,2,1)}$.

If $y = 0$ we get $\begin{cases} y = 0, \\ z = 0, \quad \text{Impossible}. \\ 0 = 1, \end{cases}$

Thus, we obtain the only point $\boxed{(0,2,1)}$. That is, the only point of the surface $\mathcal{S}$ at which the tangent plane to the surface $\mathcal{S}$ is parallel to the $xz$-plane is the point $(0,2,1)$.

4. Let $\begin{cases} D_1 := \{(x,y) \in \mathbb{R}^2\colon\ x^2 + (y+1)^2 \le 1, \quad y \le -1\} \\ D_2 := \{(x,y) \in \mathbb{R}^2\colon\ 1 \le (x-2)^2 + (y-1)^2 \le 4, \quad y \ge 1\}. \end{cases}$

(i) Show that $\displaystyle\iint_{D_1} \frac{y}{x^2+y^2}\, dxdy = -1$.

We use polar coordinates: first of all the circle $x^2 + (y+1)^2 = 1$ intersects the line $y = -1$ at the points $(-1,-1)$ which corresponds to $(\sqrt{2},\, 5\pi/4)$ and $(1,-1)$ which corresponds to $(\sqrt{2}, 7\pi/4)$. Thus

$$\begin{aligned} D_1 &= \{(x,y) \in \mathbb{R}^2\colon\ x^2 + (y+1)^2 \le 1, \quad y \le -1\} \\ &= \left\{(r,\theta) \in \mathbb{R}^2\colon\ \frac{5\pi}{4} \le \theta \le \frac{7\pi}{4}, \quad -\frac{1}{\sin\theta} \le r \le -2\sin\theta\right\}. \end{aligned}$$

Thus,

$$\begin{aligned} \iint_{D_1} \frac{y}{x^2+y^2}\, dxdy &= \int_{\frac{5\pi}{4}}^{\frac{7\pi}{4}} \int_{-\frac{1}{\sin\theta}}^{-2\sin\theta} \sin\theta dr d\theta = \int_{\frac{5\pi}{4}}^{\frac{7\pi}{4}} (1 - 2\sin^2(\theta)) d\theta \\ &= \int_{\frac{5\pi}{4}}^{\frac{7\pi}{4}} \cos(2\theta) = \left[\frac{1}{2}\sin(2\theta)\right]_{\frac{5\pi}{4}}^{\frac{7\pi}{4}} = \frac{1}{2}(-1-1) = -1. \end{aligned}$$

(ii) Show that $\displaystyle\iint_{D_2} x\,dxdy = 3\pi\,.$

Here, we use the change of variables as follows: $T:\begin{cases} x = 2 + r\cos\theta, \\ y = 1 + r\sin\theta \end{cases}$ with $\begin{cases} 0 \le \theta \le \pi, \\ 1 \le r \le 2\,. \end{cases}$

for which the Jacobian is $J_T = r$ and get

$$\begin{aligned}\iint_{D_2} x\,dxdy &= \int_0^\pi\int_1^2 (2 + r\cos\theta) r dr d\theta = \int_0^\pi \Big[r^2 + \frac{1}{3}r^3\cos\theta\Big]_1^2 d\theta \\ &= \int_0^\pi \Big[\big(4 + \frac{8}{3}\cos\theta\big) - \big(1 + \frac{1}{3}\cos\theta\big)\Big] d\theta = \int_0^\pi \big(3 + \frac{5}{3}\cos\theta\big) d\theta \\ &= \Big[3\theta + \frac{5}{3}\sin\theta\Big]_0^\pi = 3\pi\,.\end{aligned}$$

5. Let $E := \big\{(x,y,z) \in \mathbb{R}^3\colon\ x^2 + y^2 + (z+1)^2 \le 4 \text{ and } z \le 1 - x^2 - y^2\big\}$. Show that

$$\iiint_E (z+1) dxdydz \;=\; \pi\Big[\frac{21}{4} - 4\sqrt{3}\Big]\,.$$

Finding the intersection of the sphere $x^2 + y^2 + (z+1)^2 = 4$ and the paraboloid $z = 1 - x^2 - y^2$, we find the the region $E$ is enclosed by the graphs $z = -1 - \sqrt{4 - x^2 - y^2}$ and $z = 1 - x^2 - y^2$ as functions defined in the domain $D := \{(x,y) \in \mathbb{R}^2\colon\ x^2 + y^2 \le 3\}$. Hence, using cylindrical coordinates, we find that

$$\begin{aligned}\iiint_E (z+1) dxdydz &= \int_0^{2\pi}\int_0^{\sqrt{3}}\int_{-1-\sqrt{4-r^2}}^{1-r^2} (z+1) r dz dr d\theta \\ = \int_0^{2\pi}\int_0^{\sqrt{3}} r\Big[\frac{1}{2}(z+1)^2\Big]_{-1-\sqrt{4-r^2}}^{1-r^2} dr d\theta &= \frac{1}{2}\int_0^{2\pi}\int_0^{\sqrt{3}} r\big((2 - r^2)^2 - (4 - r^2)\big) dr d\theta \\ = \frac{1}{2}\int_0^{2\pi}\int_0^{\sqrt{3}} r(4 - 4r^2 + r^4 - 4 + r^2) dr d\theta &= \pi\int_0^{\sqrt{3}} \big(r^5 - 3r^3\big) dr \\ = \pi\Big[\frac{1}{6}r^6 - \frac{3}{4}r^4\Big]_0^{\sqrt{3}} &= \pi\Big(\frac{9}{2} - \frac{27}{4}\Big) = -\frac{9\pi}{4}\,.\end{aligned}$$

**Alternative approach:** We first integrate w.r.t. $xy$ and then we integrate w.r.t. $z$ as follows. If we fix $z$, we see that $(x,y)$ varies in a certain disc $D_z$ defined as:

$$D_z := \begin{cases} \{(x,y) \in \mathbb{R}^2\colon\ x^2 + y^2 \le 4 - (z+1)^2\} & \text{if } -3 \le z \le -2, \\ \{(x,y) \in \mathbb{R}^2\colon\ x^2 + y^2 \le 1 - z\} & \text{if } -2 \le z \le 1. \end{cases}$$

Thus,

$$
\begin{aligned}
\iiint_E (z+1)dxdydz &= \int_{-3}^{-2}\iint_{D_z}(z+1)dxdydz + \int_{-2}^{1}\iint_{D_z}(z+1)dxdydz \\
&= \int_{-3}^{-2}(z+1)A(D_z)dz + \int_{-2}^{1}(z+1)A(D_z)dz, \quad (A(D_z) \text{ is the area of the disc } D_z) \\
&= \int_{-3}^{-2}(z+1)\pi(4-(z+1)^2)dz + \int_{-2}^{1}(z+1)\pi(1-z)dz \\
&= \pi\Big[2(z+1)^2 - \frac{1}{4}(z+1)^4\Big]_{-3}^{-2} + \Big[z - \frac{1}{3}z^3\Big]_{-2}^{1} \\
&= \pi\Big[\big(2-\frac{1}{4}\big) - \big(8 - \frac{16}{4}\big)\Big] + \pi\Big[\big(1-\frac{1}{3}\big) - \big(-2+\frac{8}{3}\big)\Big] \\
&= \pi\big(-2-\frac{1}{4}\big) + \pi(3-3) = -\frac{9\pi}{4}\,.
\end{aligned}
$$

6. Let $\mathcal{C}$ be the closed curve described by the vector function $\vec{r}:[0,\,2\pi]\to\mathbb{R}^3$ defined by

$$\vec{r}(t) := (2\cos t + 1,\, 3\sin t) \quad \forall t \in [0, 2\pi]\,.$$

Consider the vector field $\vec{F}(x,y) := (x+y, 3y)$ for every $(x,y)\in\mathbb{R}^2$. Evaluate the line integral $\oint_{\mathcal{C}} \vec{F}\cdot d\vec{r}$ (i) directly and then (ii) using the Green theorem.

(i) A direct computation gives:

$$
\begin{aligned}
\oint_{\mathcal{C}} \vec{F}\cdot d\vec{r} &= \int_0^{2\pi} \vec{F}(\vec{r}(t))\cdot\vec{r}\,'(t)\,dt = \int_0^{2\pi}(2\cos t + 1 + 3\sin t,\, 9\sin t)\cdot(-2\sin t, 3\cos t)dt \\
&= \int_0^{2\pi}(-4\cos t\sin t - 2\sin t - 6\sin^2 t + 27\sin t\cos t)dt \\
&= \int_0^{2\pi} 23\cos t\sin t\,dt - 2\int_0^{2\pi}\sin t\,dt - 6\int_0^{2\pi}\sin^2(t)dt \\
&= \underbrace{\int_0^{2\pi}\frac{23}{2}\sin(2t)dt}_{=0} - 2\underbrace{\int_0^{2\pi}\sin t\,dt}_{=0} - 6\int_0^{2\pi}\sin^2(t)dt \\
&= -6\int_0^{2\pi}\sin^2(t)dt = 3\int_0^{2\pi}(\cos(2t)-1)dt = -6\pi\,.
\end{aligned}
$$

(ii) Using the Green theorem, we find that

$$\oint_{\mathcal{C}} \vec{F}\cdot d\vec{r} = \iint_D (0-1)dxdy = -A(D) = -\pi\times 2\times 3 = -6\pi\,.$$

where $D$ is the domain enclosed by the ellipse: $\dfrac{(x-1)^2}{2^2} + \dfrac{y^2}{3^2} = 1.$

7. Let $\mathcal{S}$ be the surface described as $\mathcal{S} := \left\{(x,y,z) \in \mathbb{R}^3\colon\ z = \dfrac{1}{\sqrt{x^2+y^2}},\quad 1 \le z \le 2\right\}$.
Show that

$$\iint_{\mathcal{S}} \frac{1}{z^4} dS = \frac{\pi}{3}\left[2\sqrt{2} - \frac{17\sqrt{17}}{64}\right].$$

Notice that $\mathcal{S}$ is the graph of the function

$$f(x,y) := \frac{1}{\sqrt{x^2+y^2}} \qquad \text{defined in the ring } D := \left\{(x,y) \in \mathbb{R}^2\colon\ \frac{1}{4} \le x^2+y^2 \le 1\right\}$$

where $\dfrac{1}{4} \le x^2+y^2 \le 1$ is obtained $\begin{cases} 1 \le z \le 2, \\ z = \dfrac{1}{\sqrt{x^2+y^2}}. \end{cases}$

Thus, using the formula of the surface integral over a graph, which is described by the vector function $\vec{r}\colon D \to \mathbb{R}^3$ defined by $\vec{r}(x,y) := (x,y,f(x,y))$ for which

$$\begin{cases} \vec{r}_x(x,y) = (1,\,0,\,f_x(x,y)), \\ \vec{r}_y(x,y) = (0,\,1,\,f_y(x,y)), \end{cases} \quad \Rightarrow \quad \vec{r}_x(x.y) \times \vec{r}_y(x,y) = (-f_x(x,y),\,-f_y(x,y),\,1)$$

$$\Rightarrow \quad \|\vec{r}_x(x.y) \times \vec{r}_y(x,y)\| = \sqrt{(f_x(x,y))^2 + (f_y(x,y))^2 + 1}\,.$$

We get that

$$\begin{aligned} \iint_{\mathcal{S}} \frac{1}{z^4} dS &= \iint_D \frac{1}{(f(x,y))^4}\,\|\vec{r}_x(x.y) \times \vec{r}_y(x,y)\|\,dxdy \\ &= \iint_D \frac{1}{(f(x,y))^4}\sqrt{(f_x(x,y))^2 + (f_y(x,y))^2 + 1}dxdy\,. \end{aligned}$$

We find that

$$\begin{cases} f_x(x,y) = -x(x^2+y^2)^{-3/2} & \Rightarrow \quad (f_x(x,y))^2 = x^2(x^2+y^2)^{-3}, \\ f_y(x,y) = -y(x^2+y^2)^{-3/2} & \Rightarrow \quad (f_y(x,y))^2 = y^2(x^2+y^2)^{-3}. \end{cases}$$

Thus

$$\begin{cases} \dfrac{1}{(f(x,y))^4} = (x^2+y^2)^2 \quad \text{and,} \\[2ex] \sqrt{f_x^2(x,y) + f_y^2(x,y) + 1} = \sqrt{(x^2+y^2)(x^2+y^2)^{-3} + 1} = \dfrac{\sqrt{(x^2+y^2)^2+1}}{x^2+y^2}. \end{cases}$$

Therefore,

$$\iint_{\mathcal{S}} \frac{1}{z^4}\,dS = \iint_D \frac{1}{(f(x,y))^4}\sqrt{f_x^2(x,y) + f_y^2(x,y) + 1}dxdy$$

$$= \iint_D (x^2+y^2)^2 \frac{\sqrt{(x^2+y^2)^2+1}}{x^2+y^2}dxdy = \underbrace{\iint_D (x^2+y^2)\sqrt{(x^2+y^2)^2+1}\,dxdy}_{\text{(To be evaluated through polar coordinates } r,\,\theta)}$$

$$= \underbrace{\int_0^{2\pi}\int_{\frac{1}{2}}^1 r^3\sqrt{r^4+1}drd\theta}_{\text{(Using polar coordinates)}} = 2\pi\left[\frac{1}{6}(r^4+1)^{3/2}\right]_{\frac{1}{2}}^1 = \frac{\pi}{3}\left[2\sqrt{2} - \frac{17\sqrt{17}}{64}\right].$$

8. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x, y) = -\sin(x - y) - \cos(2y) \quad \forall (x, y) \in \mathbb{R}^2 .$$

(i) Find **all** the critical points of the function $f$.

We seek **all** the points $(x, y) \in \mathbb{R}^2$ such that $f_x(x, y) = f_y(x, y) = 0$. We find that

$$f_x(x, y) = -\cos(x - y) \quad \text{while} \quad f_y(x, y) = \cos(x - y) + 2\sin(2y) .$$

Hence, we obtain the system of equations

$$\begin{cases} -\cos(x - y) = 0, \\ \cos(x - y) + 2\sin(2y) = 0 \end{cases} \Leftrightarrow \begin{cases} \cos(x - y) = 0, \\ \sin(2y) = 0 \end{cases} \Leftrightarrow \begin{cases} x - y = \dfrac{\pi}{2} + k\pi \quad k \in \mathbb{Z}, \\ 2y = n\pi, \quad n \in \mathbb{Z}, \end{cases}$$

So, we find the points

$$\boxed{\Big((n + 1)\frac{\pi}{2} + k\pi, \ \frac{n\pi}{2}\Big) \quad k, n \in \mathbb{Z}} .$$

(ii) Determine the nature (i.e., point of local maximum/minimum or saddle point) of each of the points $\big(\pi, \frac{\pi}{2}\big)$ and $\big(\frac{7\pi}{2}, \pi\big)$.

We will apply the second order optimality criteria for which we will need the second order partial derivatives:

$$f_{xx}(x, y) = \sin(x - y), \quad f_{xy}(x, y) = -\sin(x - y), \quad f_{yy} = \sin(x - y) + 4\cos(2y) .$$

Now,

$$D(a, b) = \begin{vmatrix} f_{xx}\big(\pi, \frac{\pi}{2}\big) & f_{xy}\big(\pi, \frac{\pi}{2}\big) \\ f_{xy}\big(\pi, \frac{\pi}{2}\big) & f_{yy}\big(\pi, \frac{\pi}{2}\big) \end{vmatrix} = \begin{vmatrix} 1 & -1 \\ -1 & -3 \end{vmatrix} = -4 < 0$$

and hence, $\big(\pi, \frac{\pi}{2}\big)$ is a saddle point.
On the other hand,

$$\begin{vmatrix} f_{xx}\big(\frac{7\pi}{2}, \pi\big) & f_{xy}\big(\frac{7\pi}{2}, \pi\big) \\ f_{xy}\big(\frac{7\pi}{2}, \pi\big) & f_{yy}\big(\frac{7\pi}{2}, \pi\big) \end{vmatrix} = \begin{vmatrix} 1 & -1 \\ -1 & 4 \end{vmatrix} = 3 .$$

So, $D\big(\frac{7\pi}{2}, \pi\big) > 0$ and $f_{xx}\big(\frac{7\pi}{2}, \pi\big) > 0$ (or $f_{yy}\big(\frac{7\pi}{2}, \pi\big) > 0$). Thus, $\big(\frac{7\pi}{2}, \pi\big)$ is a point of local minimum.

# ALL THE BEST!